

Lecture 13: Number Systems (Binary, Decimal, Octal & Hexadecimal)

Learning Objectives

By the end of this lecture, students will be able to:

- Understand what a number system is.
 - Explain why computers use binary.
 - Identify Decimal, Binary, Octal, and Hexadecimal number systems.
 - Understand the concept of base (radix).
 - Learn place values in different number systems.
 - Convert numbers between Binary, Decimal, Octal, and Hexadecimal.
 - Avoid common mistakes in number system conversions.
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1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally include:

"In today's Computer Organization and Assembly Language lecture, we'll learn Number Systems, including Binary, Decimal, Octal, and Hexadecimal, understand why computers use Binary, explore place values, and prepare for number system conversions with easy examples."

In the previous lecture, we learned about Instruction Set Architecture (ISA), RISC, and CISC. Now we're moving into one of the most important topics in Computer Organization—**Number Systems**.

Before understanding Assembly Language, memory addresses, machine instructions, and digital circuits, we first need to understand how computers represent numbers.

As humans, we normally use the Decimal Number System because we have ten fingers. However, computers are electronic devices that operate using only two electrical states: ON and OFF.

Because of this, computers use the **Binary Number System**.

In today's lecture, we'll study the four most important number systems used in computing: Decimal, Binary, Octal, and Hexadecimal. We'll also learn how to convert between them using simple step-by-step methods.

Let's begin.

2. What is a Number System?

Definition

A Number System is a method of representing numbers using a specific set of digits and a base (radix).

In simple words:

A number system defines **which digits can be used** and **how the value of a number is calculated**.

Examples:

- Decimal uses digits **0–9**.
 - Binary uses digits **0 and 1**.
 - Octal uses digits **0–7**.
 - Hexadecimal uses digits **0–9 and A–F**.
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3. What is Base (Radix)?

The **base** of a number system is the number of unique symbols (digits) it uses.

Number System	Base	Digits Used
Decimal	10	0–9
Binary	2	0, 1
Octal	8	0–7
Hexadecimal	16	0–9, A–F

Examples:

- $(25)_{10} \rightarrow \text{Decimal}$
- $(1011)_2 \rightarrow \text{Binary}$
- $(347)_8 \rightarrow \text{Octal}$
- $(2F)_{16} \rightarrow \text{Hexadecimal}$

4. Why Do Computers Use Binary?

Computers are built using electronic circuits and transistors.

A transistor has two stable states:

OFF $\rightarrow$ 0

ON $\rightarrow$ 1

Because there are only two reliable states, computers naturally use **Binary**.

Advantages of Binary:

- Easy to represent electronically.
 - Less affected by electrical noise.
 - Reliable and efficient.
 - Simplifies digital circuit design.
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5. Decimal Number System (Base 10)

This is the number system we use every day.

Digits:

0 1 2 3 4 5 6 7 8 9

Place Values

Example:

583

Expand it:

$$5 \times 10^2 + 8 \times 10^1 + 3 \times 10^0$$

Calculate:

$$5 \times 100 = 500$$

$$8 \times 10 = 80$$

$$3 \times 1 = 3$$

Final Answer:

583

6. Binary Number System (Base 2)

Binary uses only two digits:

0

1

Place Values

$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

Example:

$$101101_2$$

Expand:

$$1 \times 32 + 0 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$$

Calculate:

$$32 + 8 + 4 + 1 = 45$$

Therefore:

$$101101_2 = 45_{10}$$

7. Octal Number System (Base 8)

Digits used:

0 1 2 3 4 5 6 7

Place Values:

$$8^0 = 1$$

$$8^1 = 8$$

$$8^2 = 64$$

$$8^3 = 512$$

Example:

$$357_8$$

Expand:

$$3 \times 64 + 5 \times 8 + 7 \times 1$$

Calculate:

$$192 + 40 + 7 = 239$$

So:

$$357_8 = 239_{10}$$

8. Hexadecimal Number System (Base 16)

Digits:

0 1 2 3 4 5 6 7 8 9

A B C D E F

Letters represent:

A = 10

B = 11

C = 12

D = 13

E = 14

F = 15

Place Values:

$$16^0 = 1$$

$$16^1 = 16$$

$$16^2 = 256$$

Example:

$$2F_{16}$$

Expand:

$$2 \times 16 + 15 \times 1$$

Calculate:

$$32 + 15 = 47$$

Therefore:

$$2F_{16} = 47_{10}$$

9. Comparing Number Systems

Decimal	Binary	Octal	Hexadecimal
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0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
15	1111	17	F

16	10000	20	10
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10. Number System Conversion Overview

In the next lectures, we'll learn detailed conversion methods.

For now, remember:

- Binary ↔ Decimal
- Decimal ↔ Binary
- Binary ↔ Octal
- Binary ↔ Hexadecimal
- Octal ↔ Hexadecimal (through Binary)

These conversions are essential for assembly language and computer architecture.

11. Real-Life Analogy

Imagine different languages.

- English
- Urdu
- Arabic
- Chinese

The **meaning** remains the same, but the **representation** changes.

Similarly:

25_{10}

11001_2

31_8

19_{16}

All represent the **same value (25)**, just written in different number systems.

12. Practical Examples

Example 1

Decimal:

18_{10}

Binary:

10010_2

Example 2

Decimal:

15_{10}

Hexadecimal:

F_{16}

Example 3

Decimal:

8_{10}

Octal:

10_8

13. Common Mistakes

Students often:

- Forget the base of the number.
- Use digits that are not allowed (e.g., 9 in Octal or 2 in Binary).
- Ignore place values.
- Forget that A–F represent 10–15 in Hexadecimal.

Encourage students to always write the base as a subscript, such as $(1010)_2$ or $(2F)_{16}$.



Lecture Summary

Today, we introduced Number Systems and learned why they are fundamental to Computer Organization and Assembly Language.

We explored the Decimal, Binary, Octal, and Hexadecimal number systems, understood the concept of base (radix), studied place values, and saw how the same value can be represented in different number systems.

This knowledge is essential because processors, memory addresses, and machine instructions are all based on binary representation.

In the next lecture, we'll begin **Number System Conversions**, where we'll learn step-by-step methods to convert numbers between Binary, Decimal, Octal, and Hexadecimal with plenty of solved examples.

Thank you for watching, and I'll see you in the next lecture.
